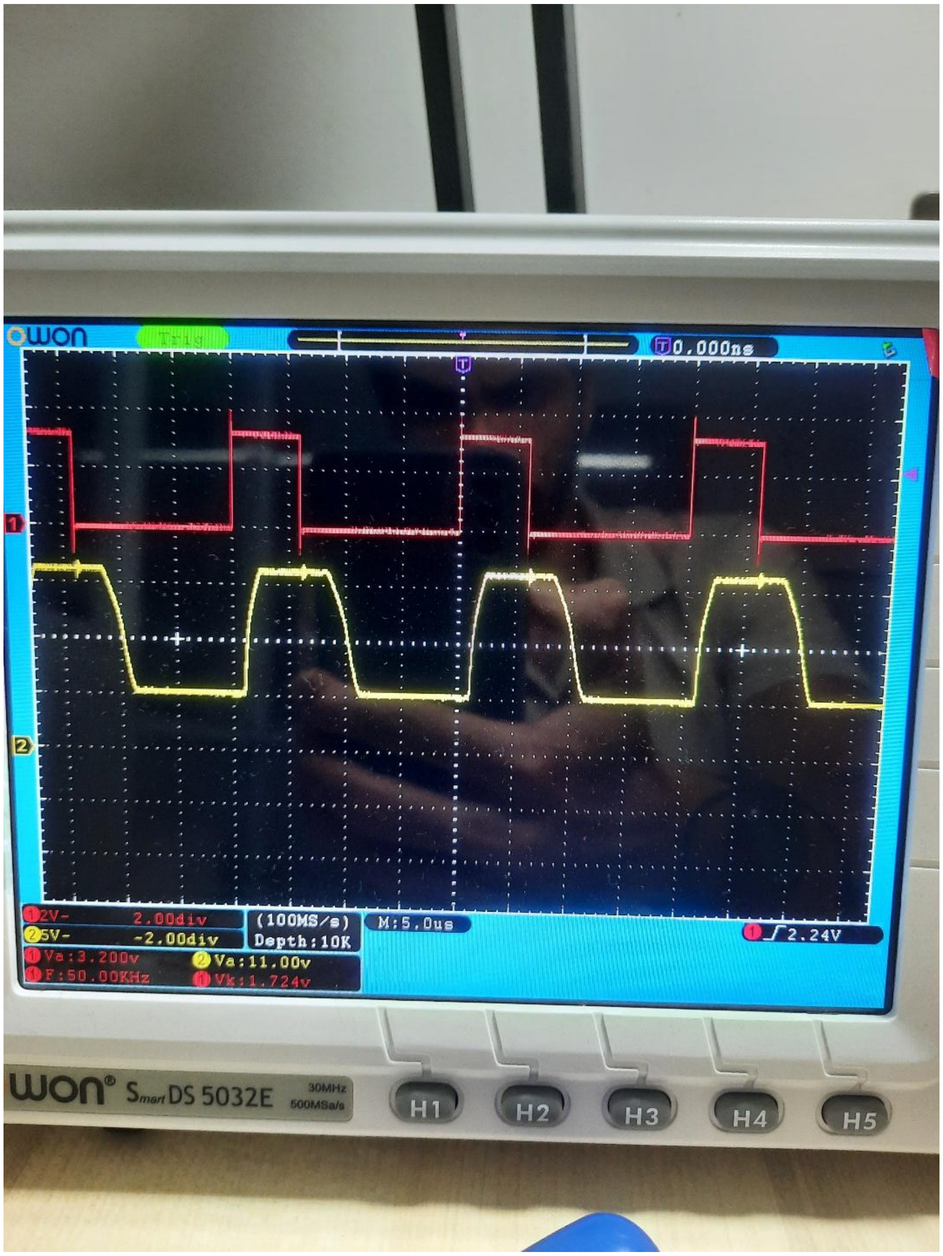
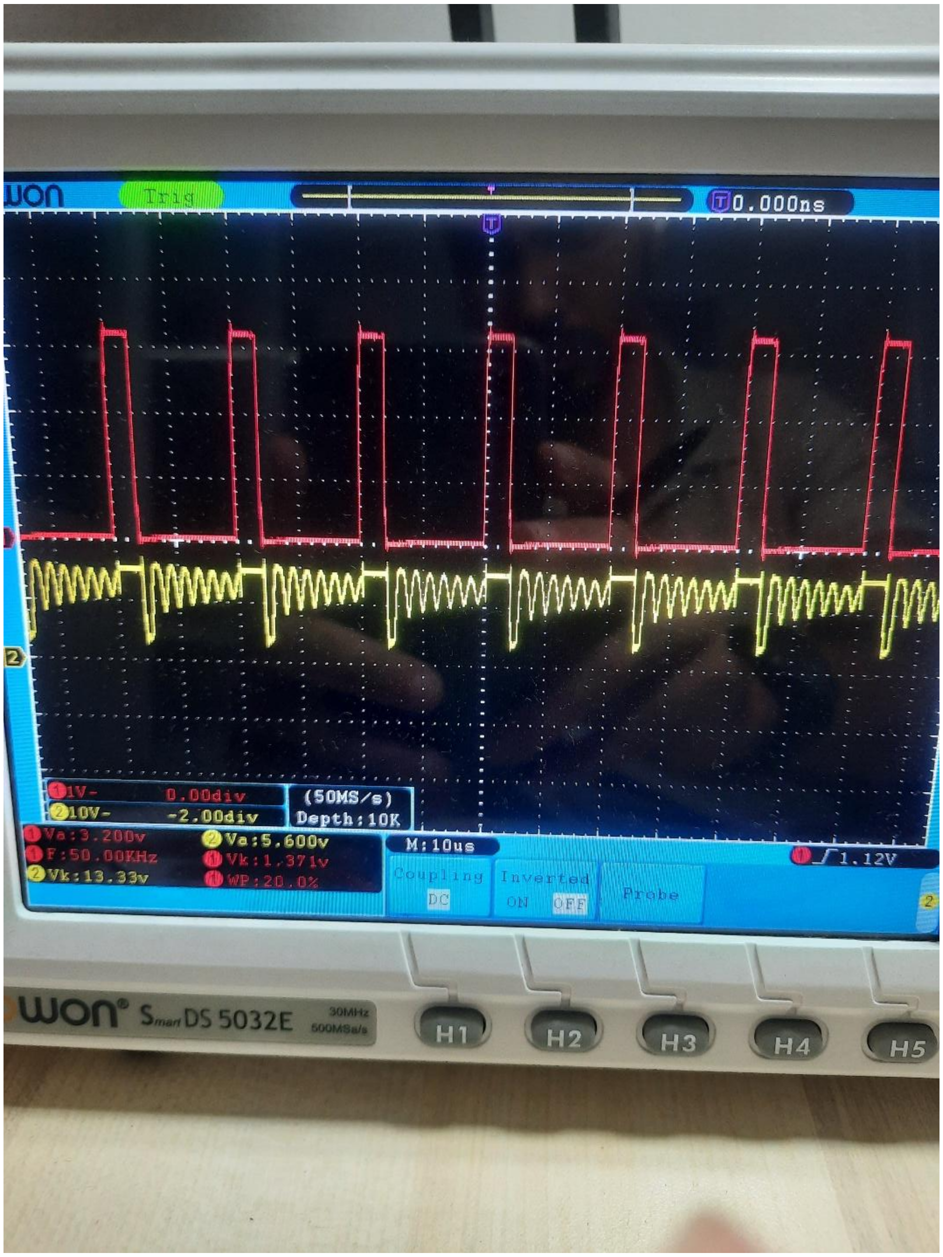
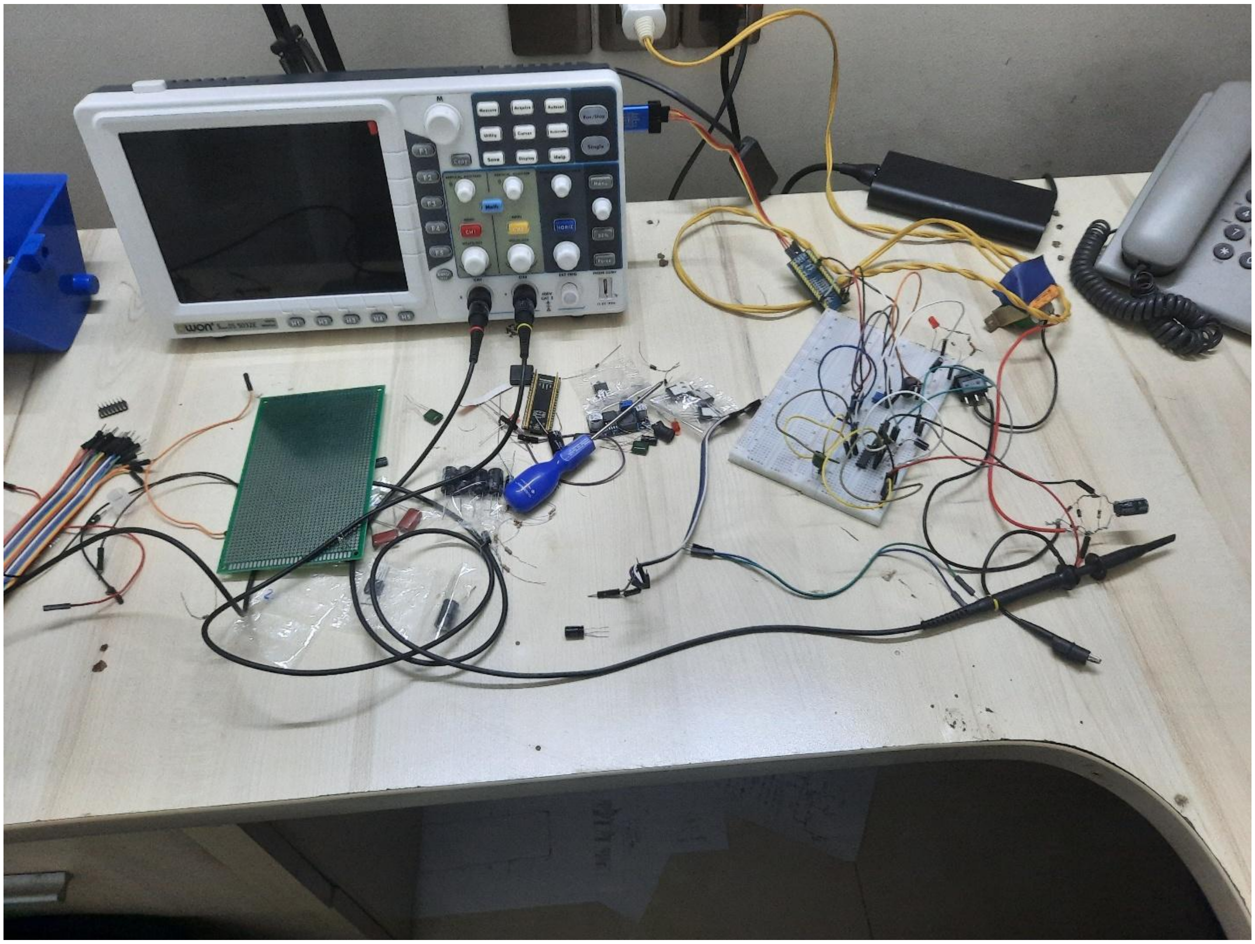
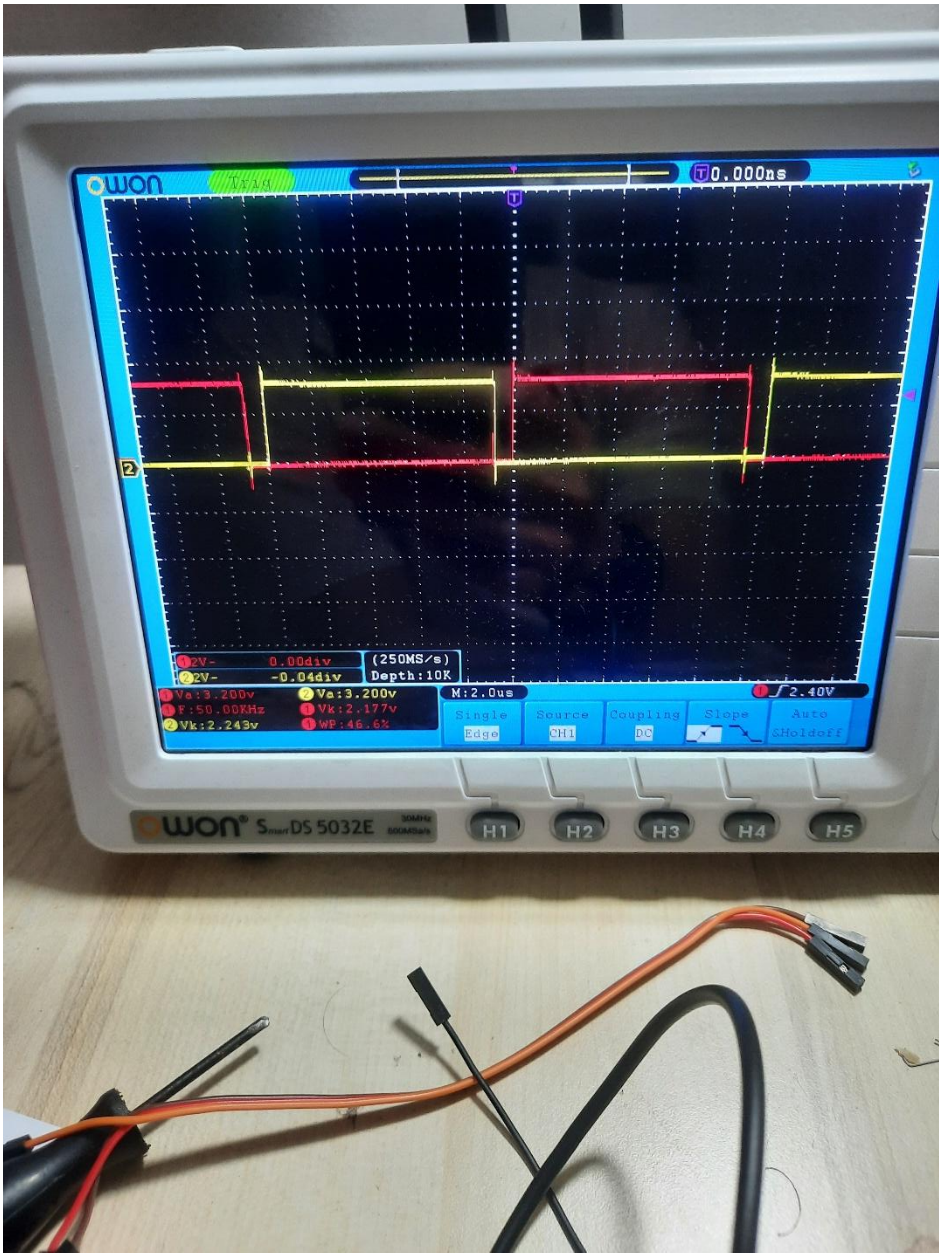
VFD Control System: Development Portfolio

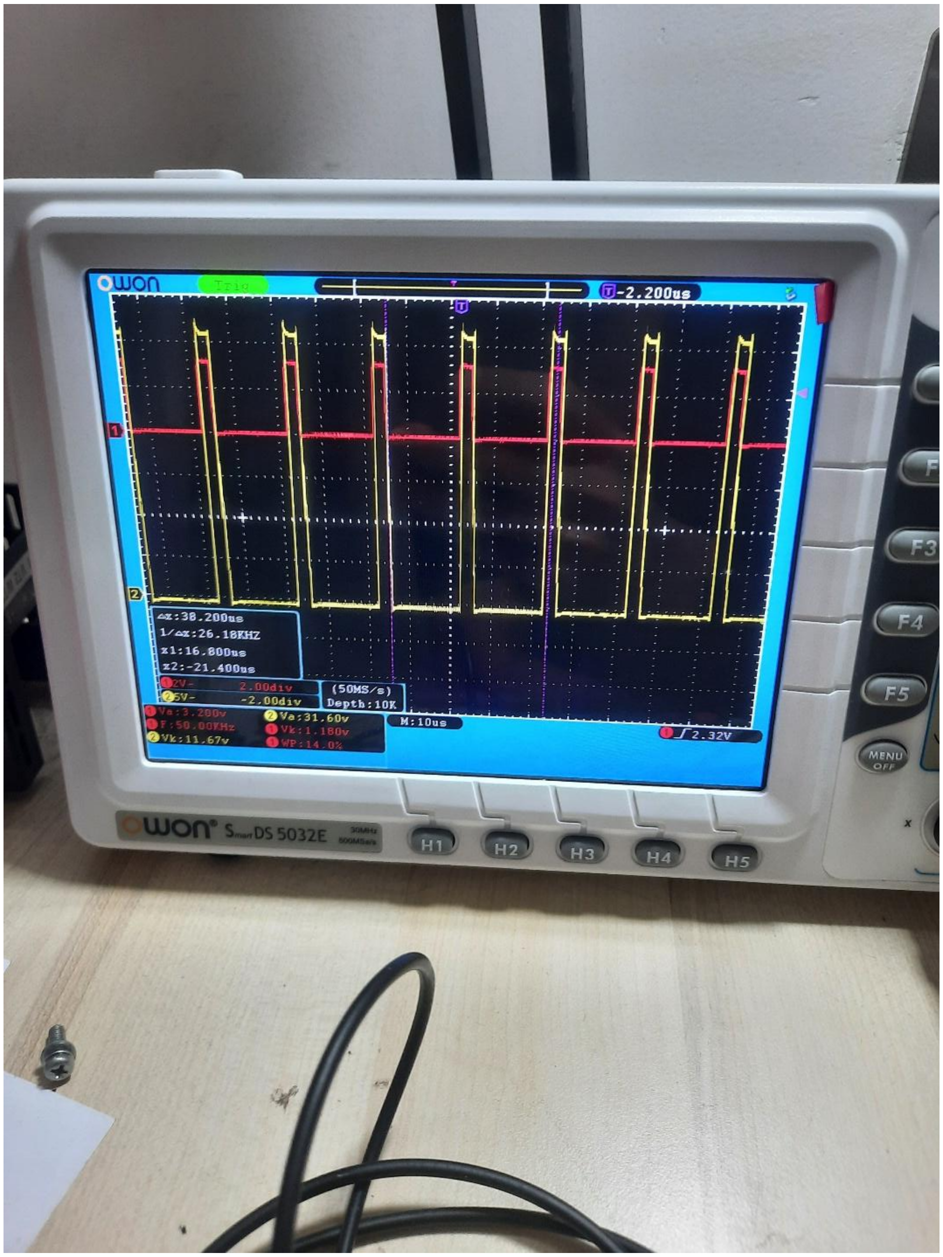




Trig
0.000ns
1V- 0.00div
10V- -2.00div
(50MS/s)
Depth:10K
Va:3.200v
F:50.00KHz
Vk:13.33v
Va:5.600v
Vk:1.371v
WP:20.0%
M:10us
1.12V
Coupling
DC
Inverted
ON OFF
Probe
Smart DS 5032E
30MHz
500MSa/s
H1
H2
H3
H4
H5




OWON
Trig
0.000ns
2V- 0.00div
2V- -0.04div
(250MS/s)
Depth:10K
Va:3.200v
F:50.00KHz
Vk:2.243v
Va:3.200v
Vk:2.177v
WP:46.6%
M:2.0us
2.40V
Single
Edge
Source
CH1
Coupling
DC
Slope
Auto
&Holdoff
OWON SmartDS 5032E
H1
H2
H3
H4
H5


-2.200us
Δx:38.200us
1/Δx:26.18KHZ
x1:16.800us
x2:-21.400us
2V-
2.00div
5V-
-2.00div
(50MS/s)
Depth:10K
Va:3.200v
F:50.00KHz
Vk:11.67v
Va:31.60v
Vk:1.180v
WP:14.0%
M:10us
2.32V
F3
F4
F5
MENU
OFF
OWON SmartDS 5032E
H1
H2
H3
H4
H5

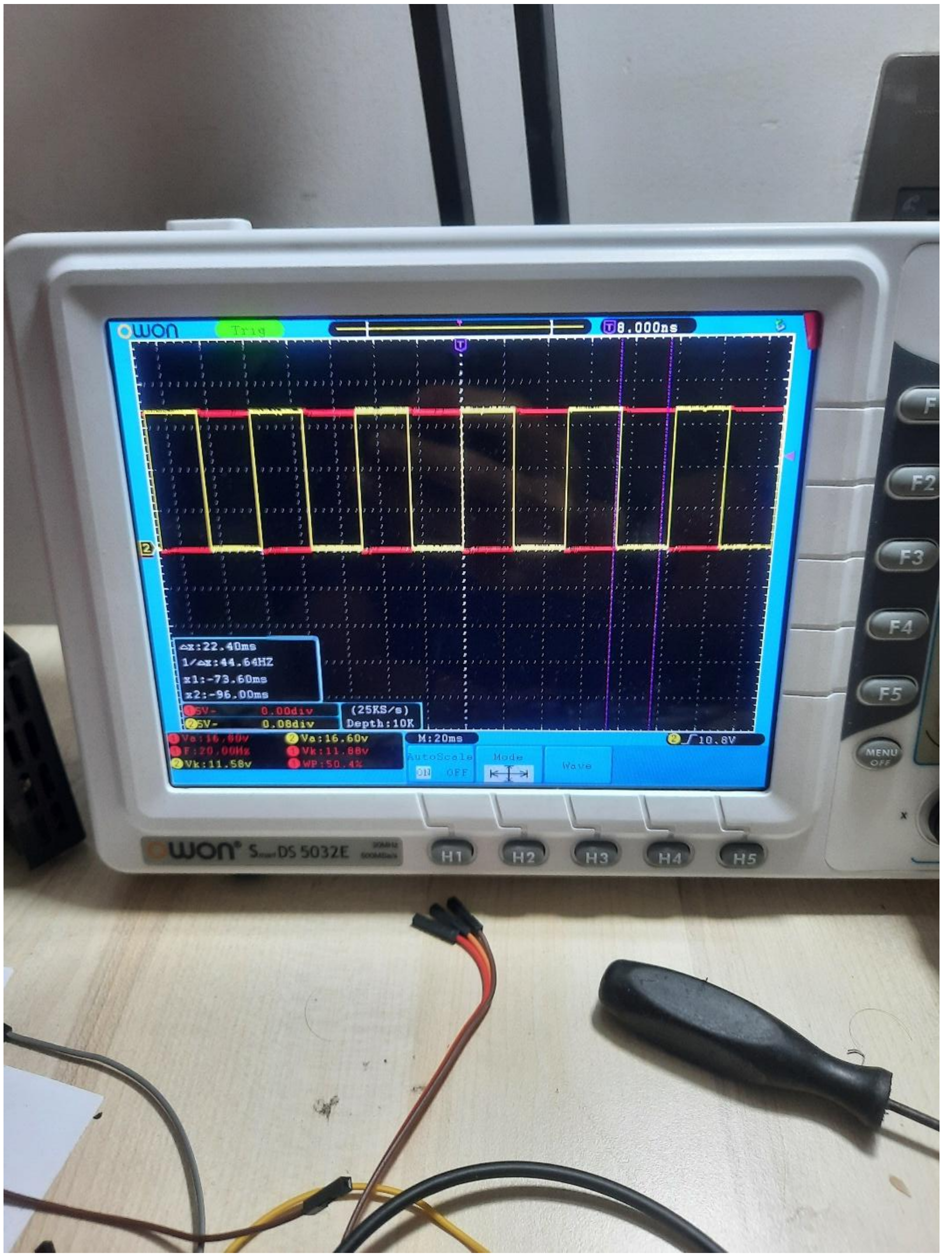
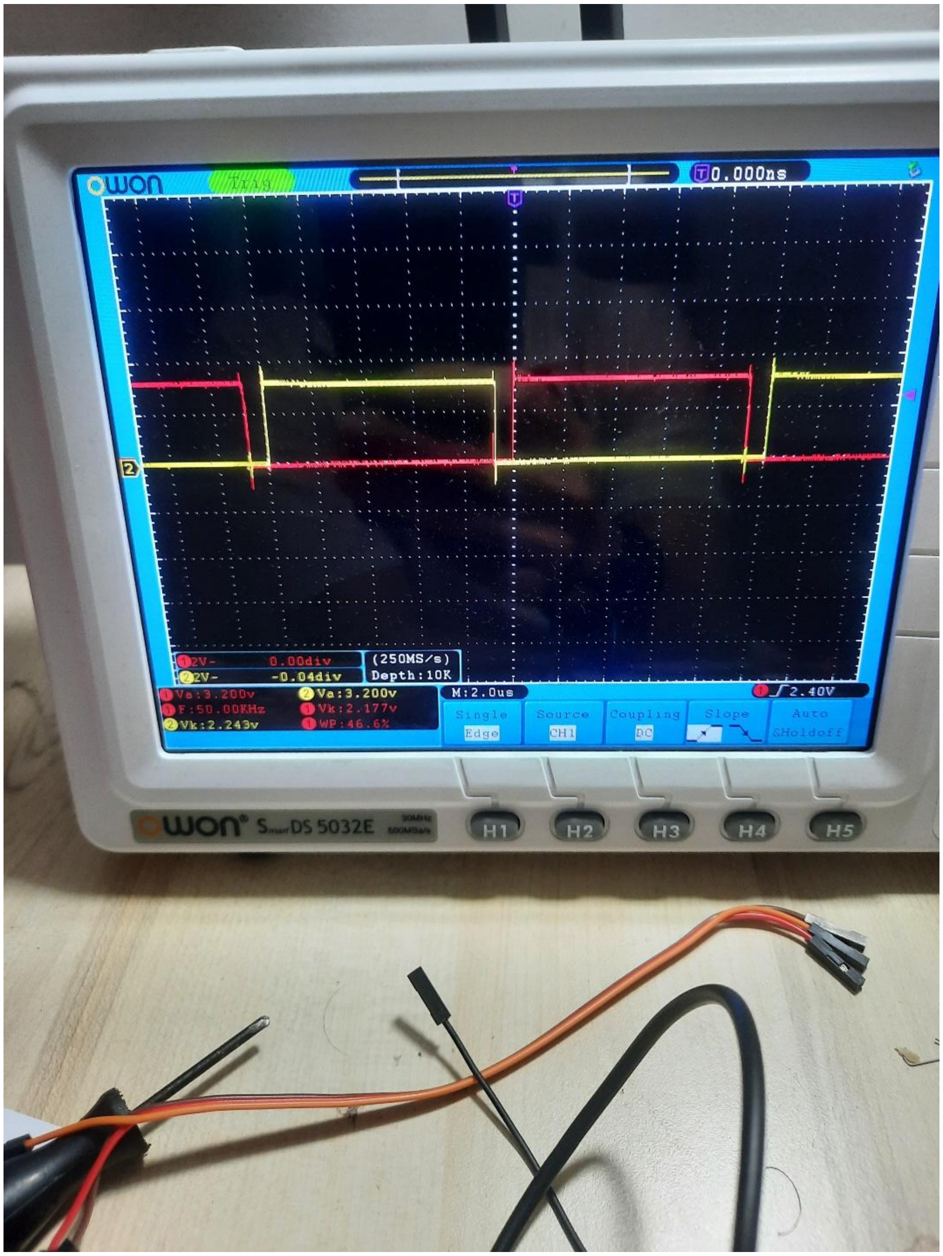
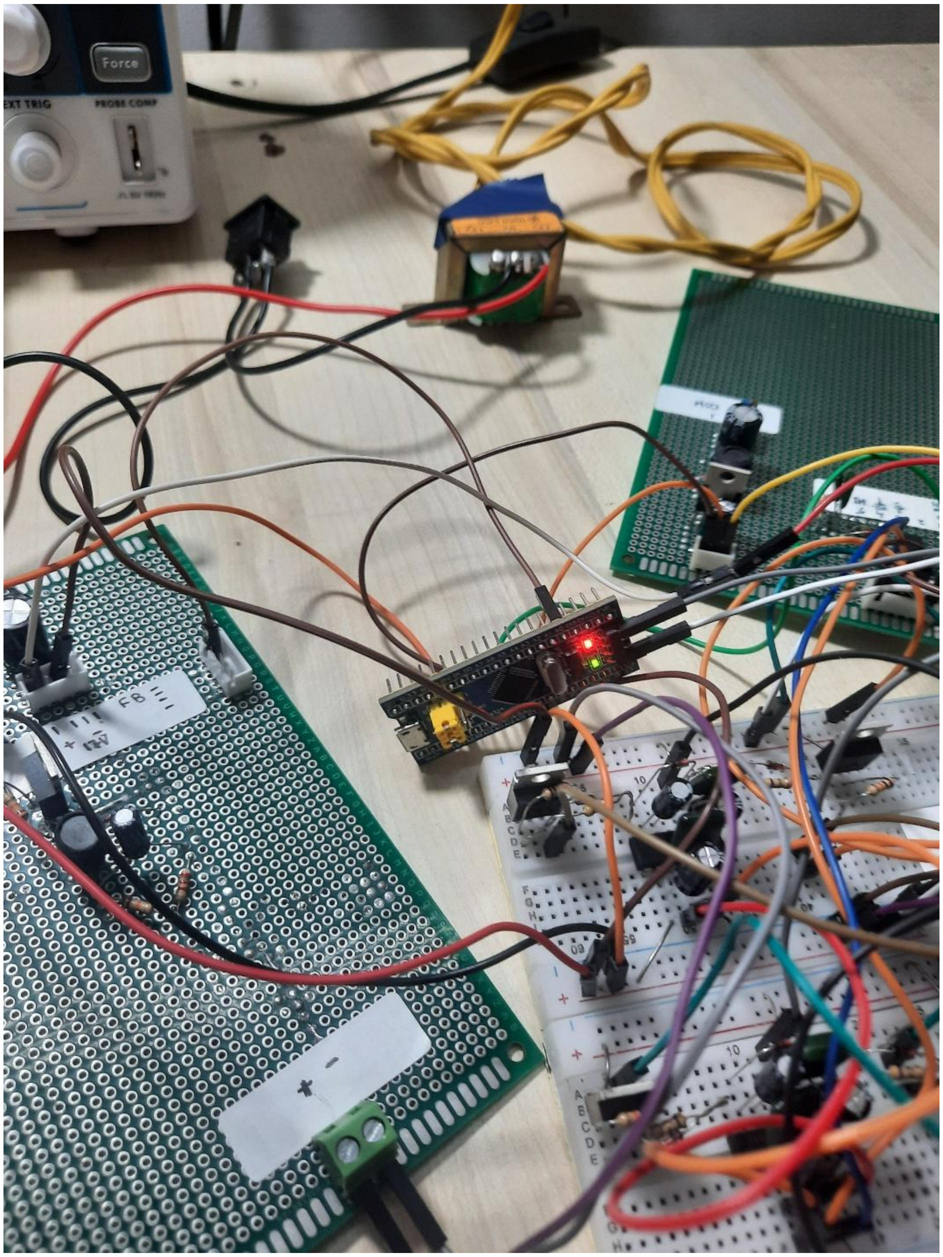
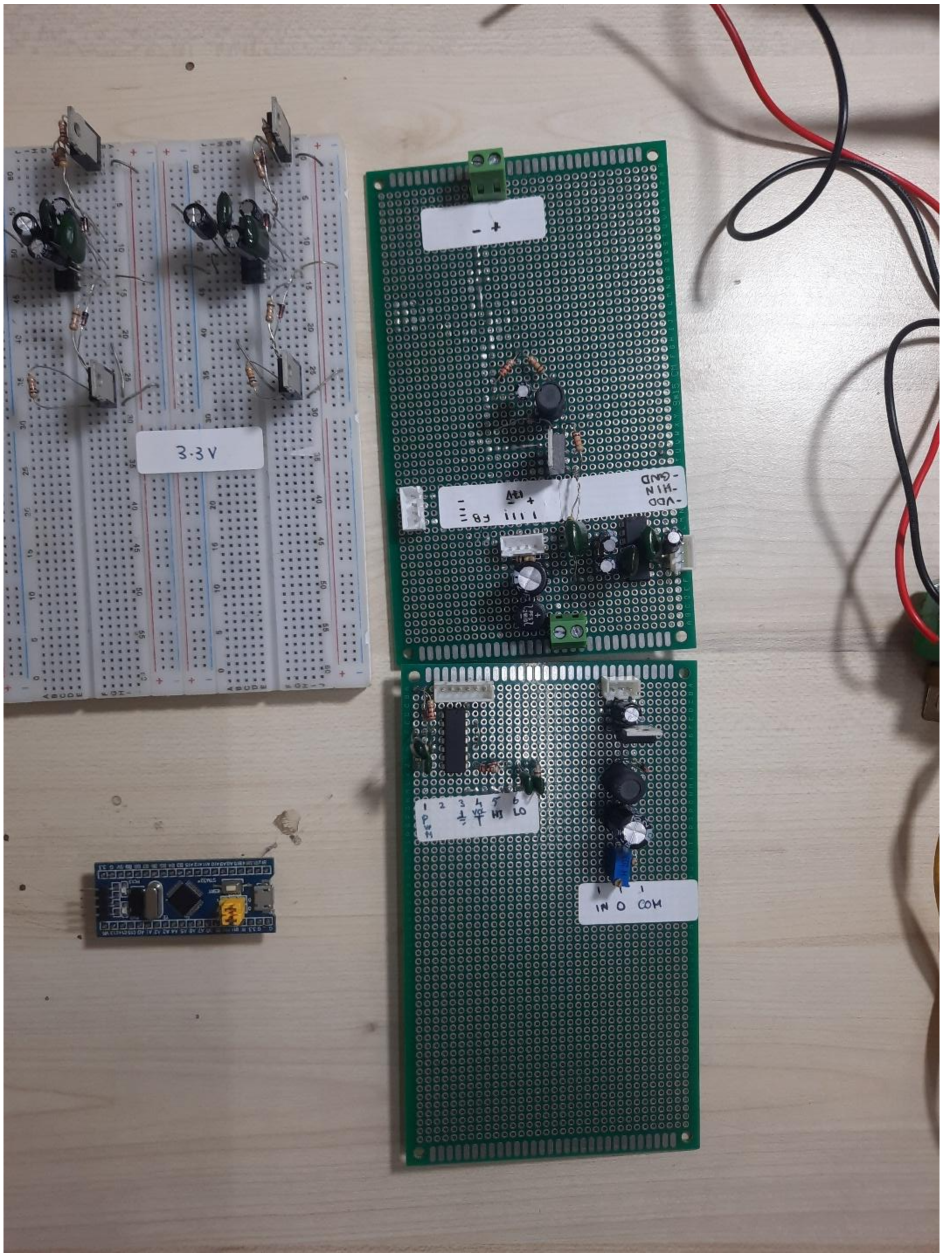
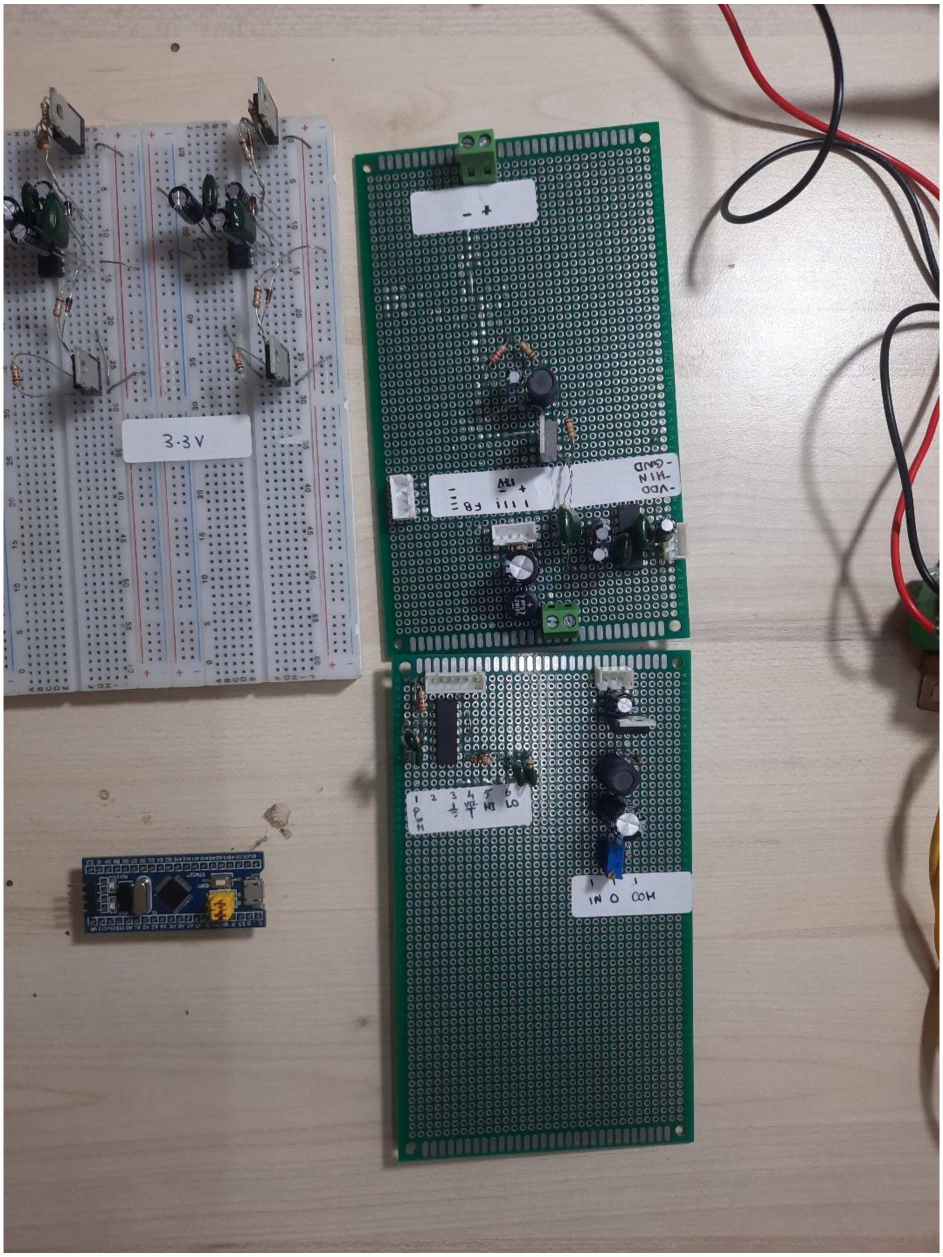


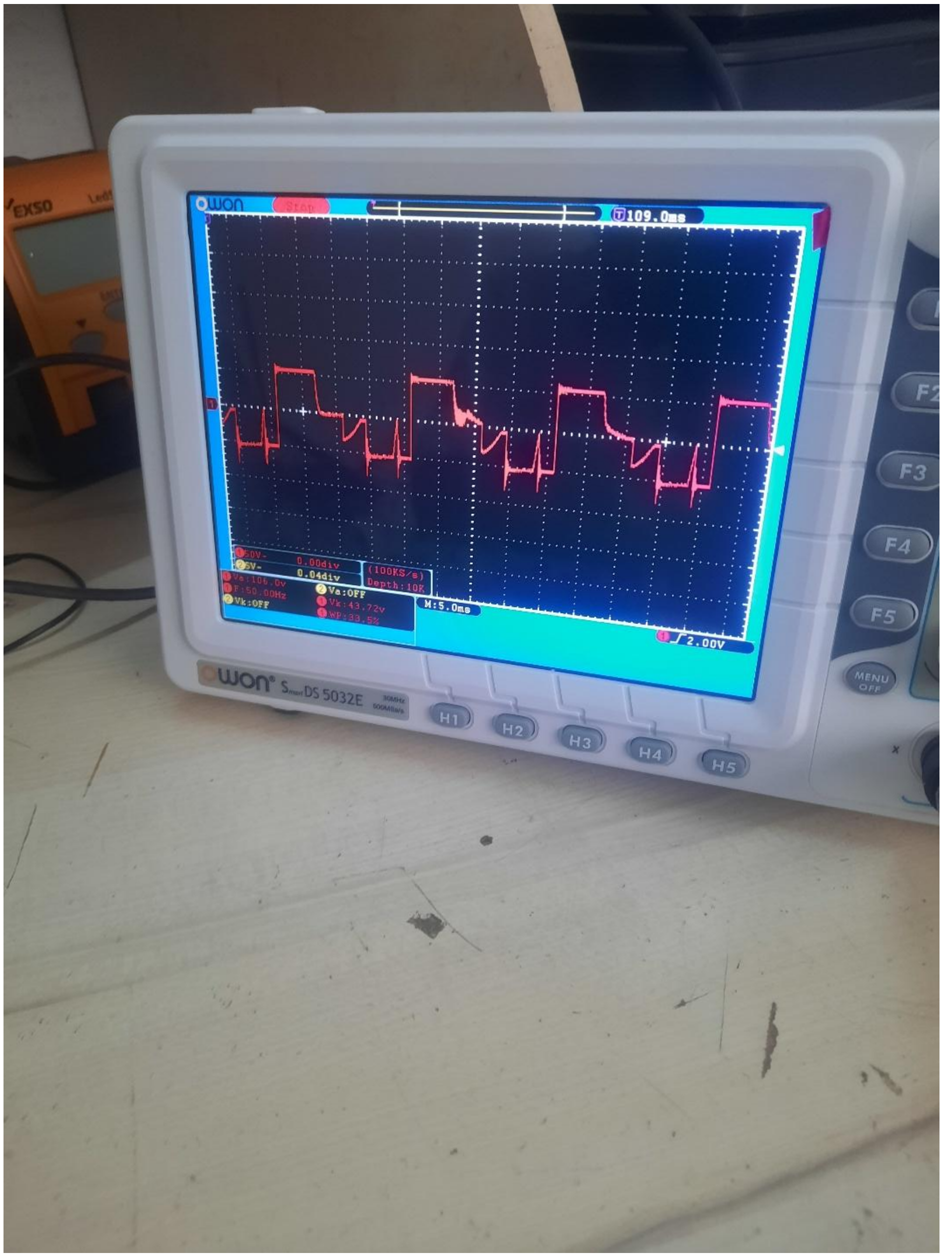
Figure 1Deadtime generation circuit testing


Trig
0.000ns
2V- 0.00div
2V- -0.04div
(250MS/s)
Depth:10K
Va:3.200v
Va:3.200v
F:50.00KHz
Vk:2.177v
Vk:2.243v
WP:46.6%
M:2.0us
2.40V
Single
Edge
Source
CH1
Coupling
DC
Slope
Auto
&Holdoff
DS 5032E
H1
H2
H3
H4
H5

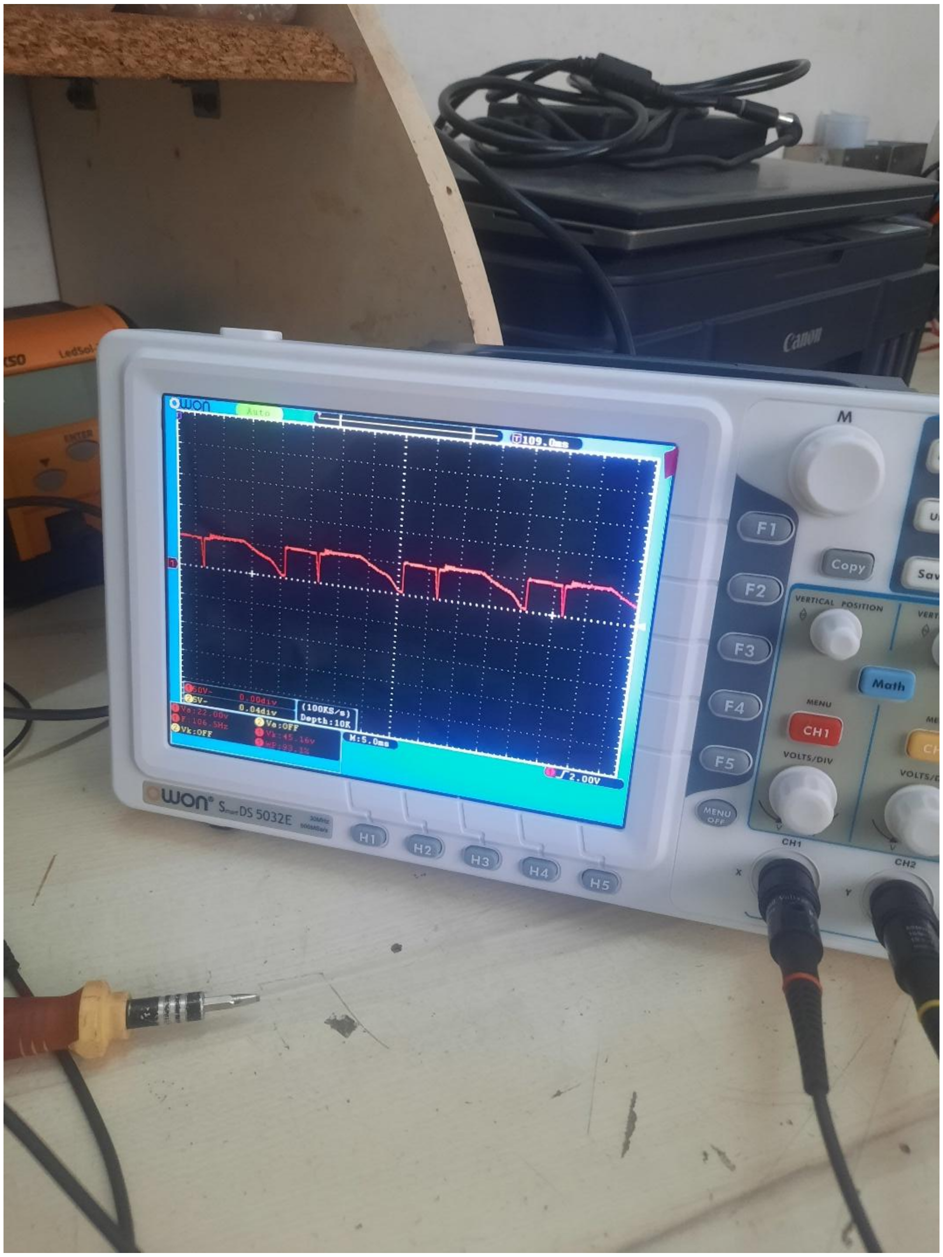

Force
EXT TRIG
PROBE COMP
FB


3.3V
- +
IN O COM


3.3V
IN O COM



109.0ms
50V~
0.00div
5V~
0.04div
(100KS/s)
Depth:10K
Va:106.0v
F:50.00Hz
Vk:OFF
Va:OFF
Vk:43.72v
WP:33.5%
M:5.0ms
2.00V
OWON
SmartDS 5032E
F2
F3
F4
F5
MENU
OFF
H1
H2
H3
H4
H5



Auto
109.0ms
M:5.0ms
2.00V
M
F1
F2
F3
F4
F5
Copy
Math
CH1
VOLTS/DIV
CH1
CH2
H1
H2
H3
H4
H5
OWON
DS 5032E
Canon

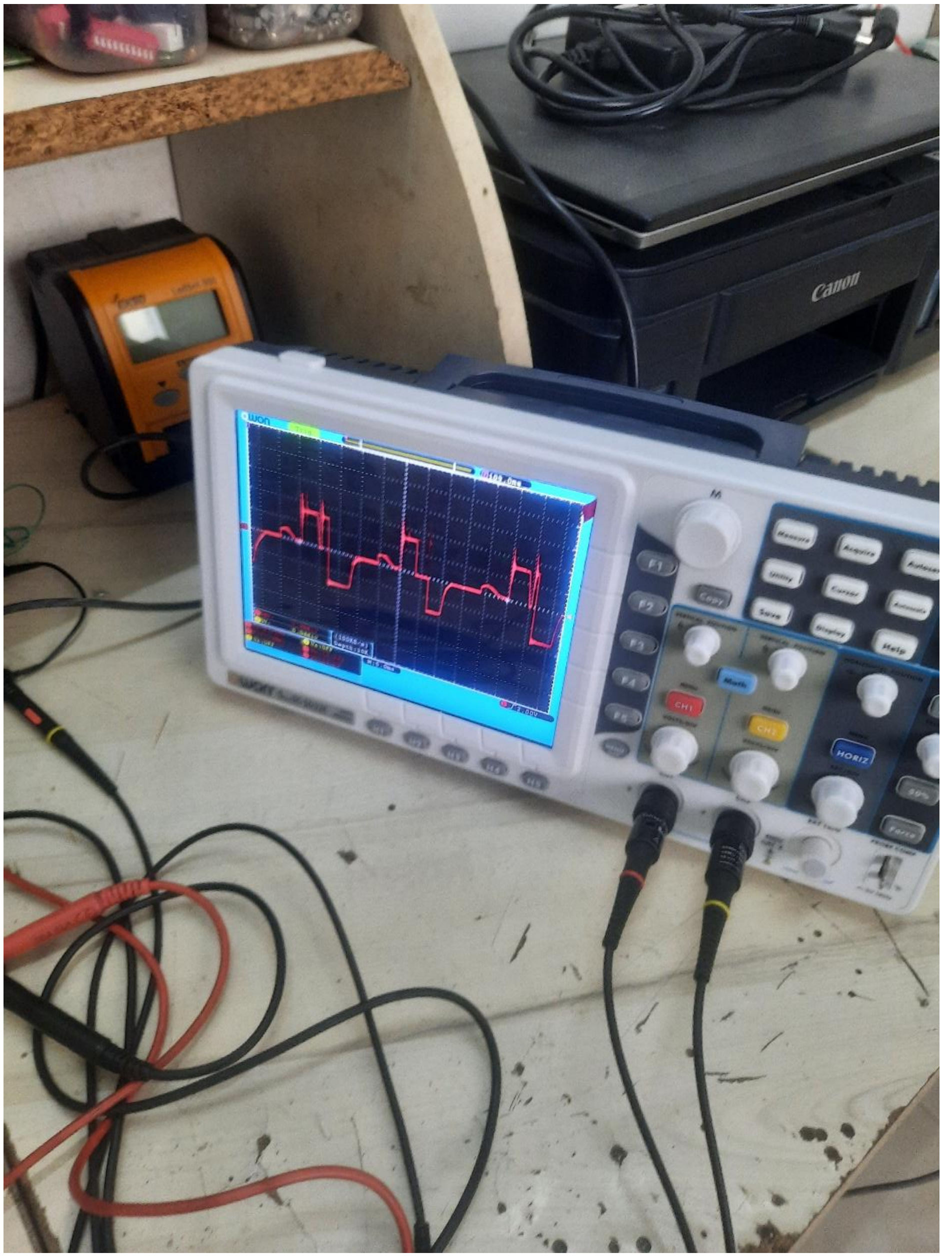
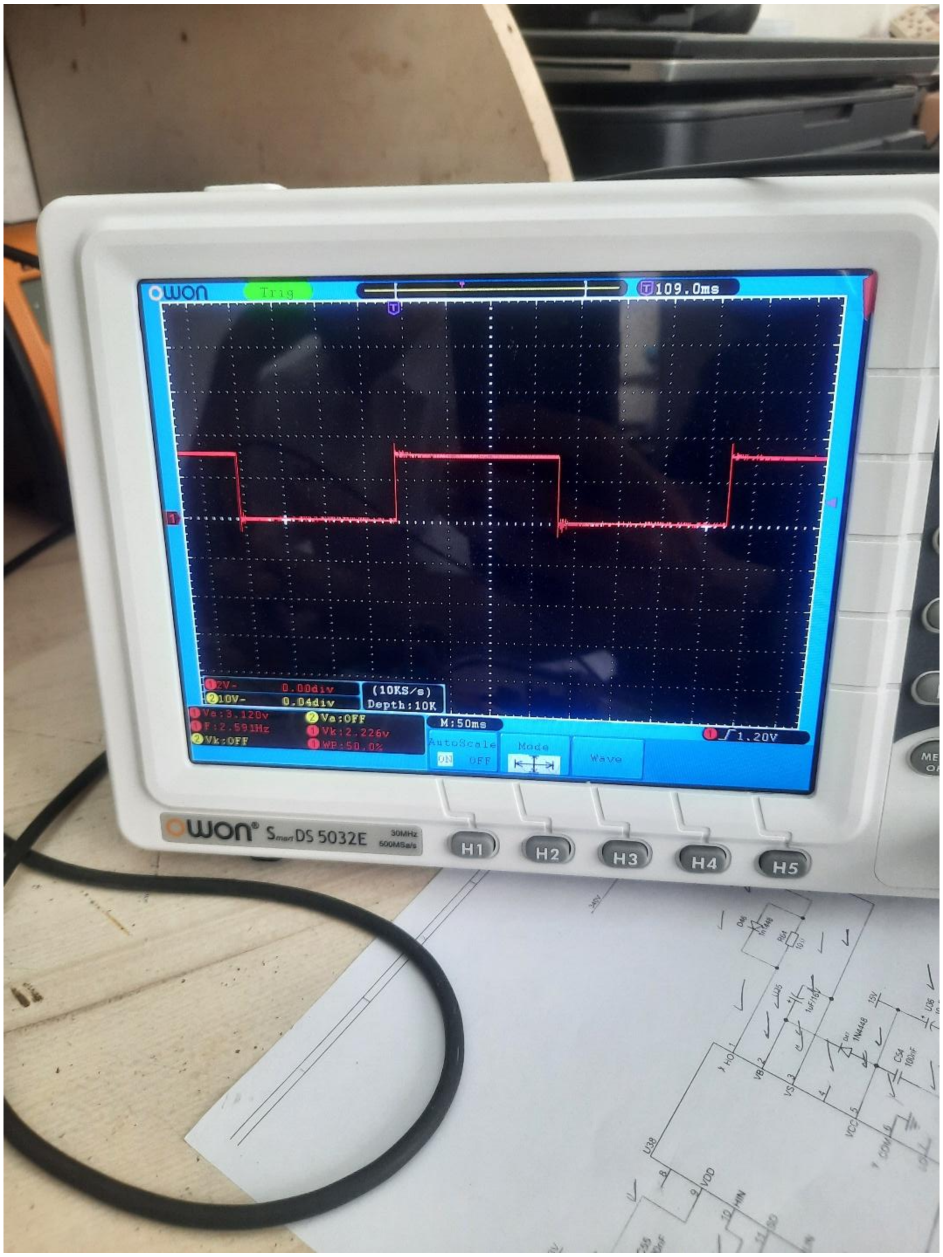


Figure 2First working prototype with the vibrator

Figure 3 High voltage buck converter



Trig
109.0ms
2V~ 0.00div
10V~ 0.04div
(10KS/s)
Depth:10K
Vs:3.120v
F:2.591Hz
Vk:OFF
Va:OFF
Vk:2.226v
WP:50.0%
M:50ms
AutoScale
ON OFF
Mode
Wave
1.20V
OWON SmartDS 5032E
30MHz
600MSa/s
H1
H2
H3
H4
H5

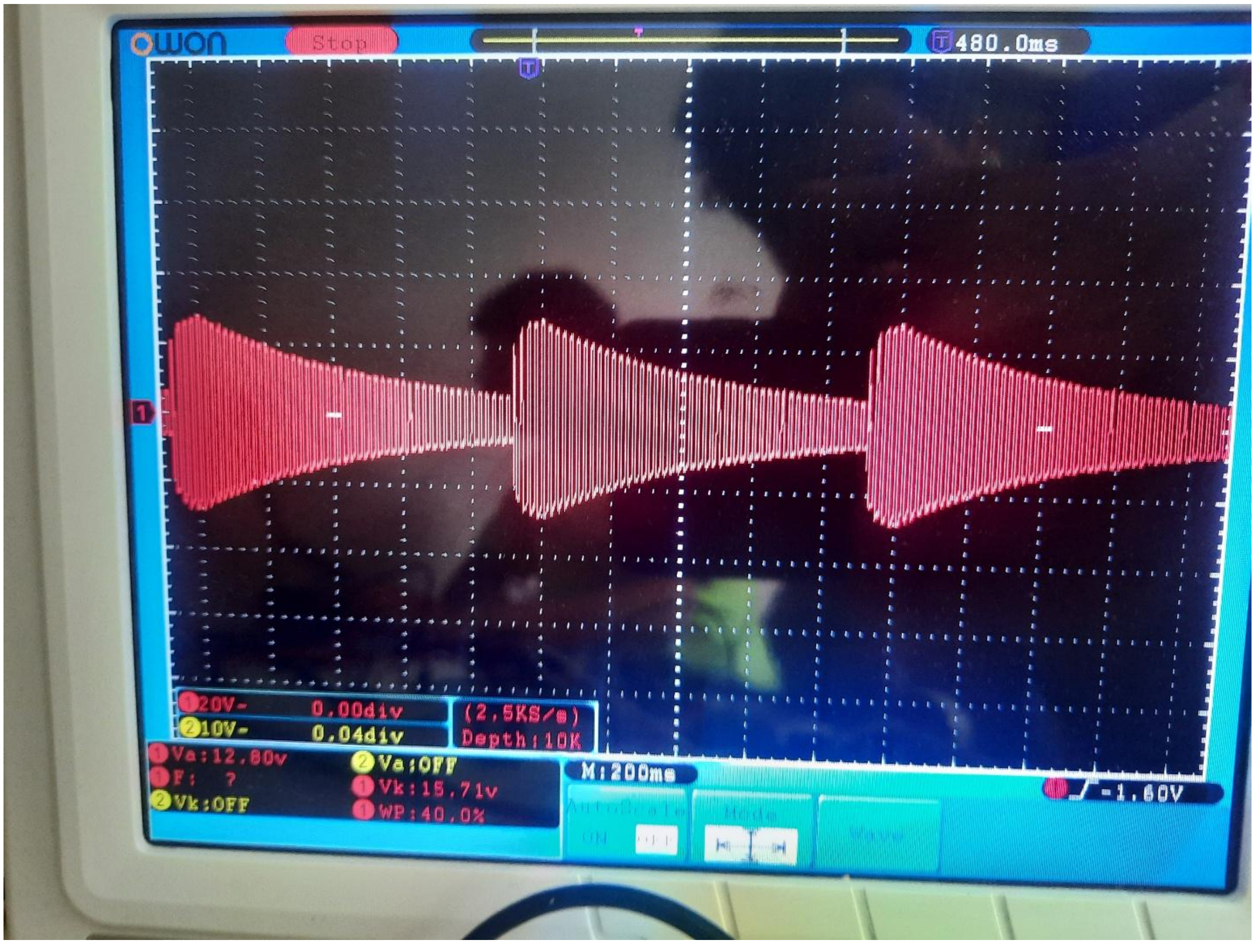


Figure 4 impedance mismatch between the VFD and the motor

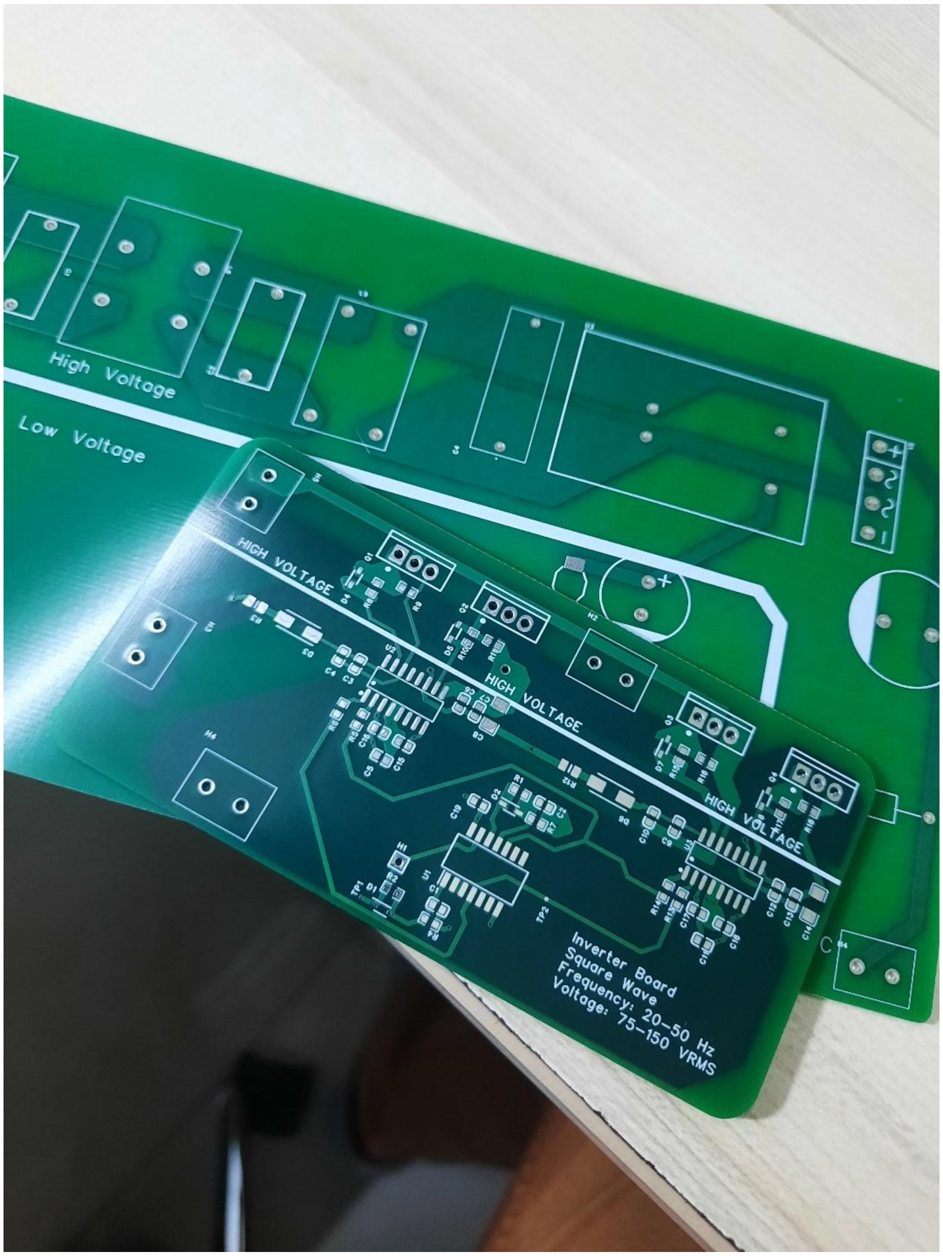


Figure 5 PCB shipment arrives

Buck Converter Stage

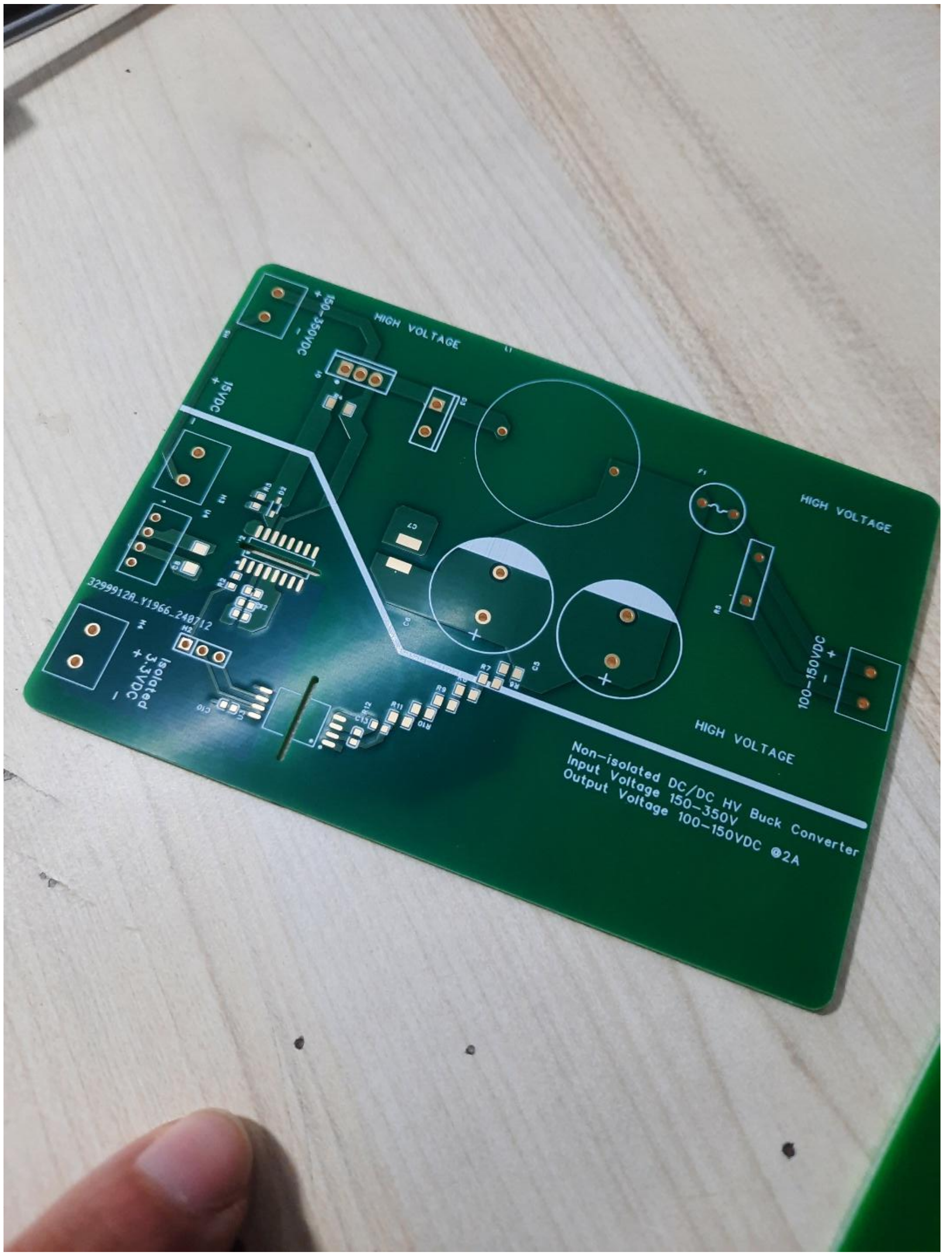


Figure 6 buck stage PCB

Figure 7 assembled AC-DC circuit with filers and power supply


HIECUBE
HE15P24LRN
High Voltage
Low Voltage
AC-DC Converter & Power Supply
Vout: 15V and Isolated 3.3V
340VDC
15V
3.3V
AC

Buck Converter Stage

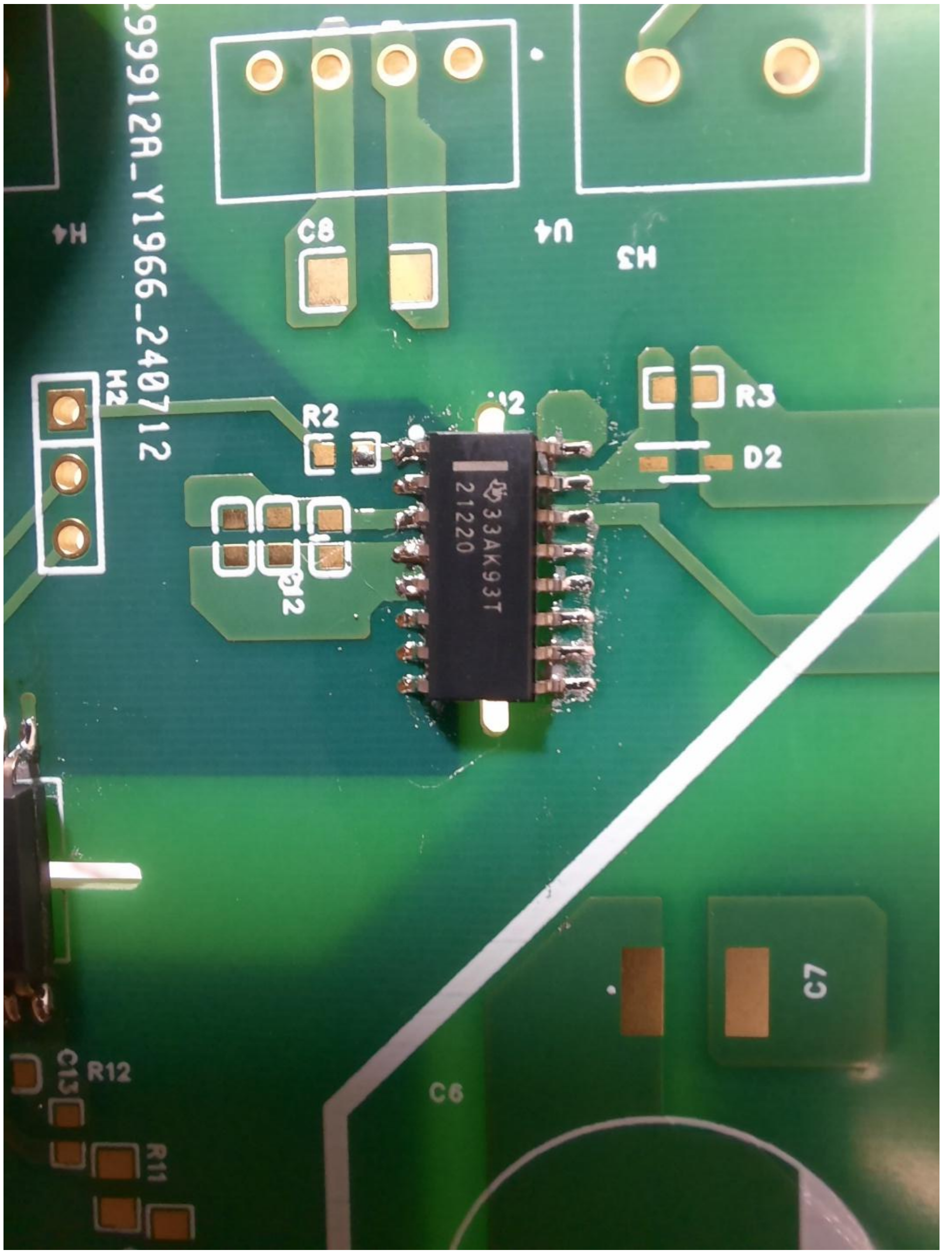
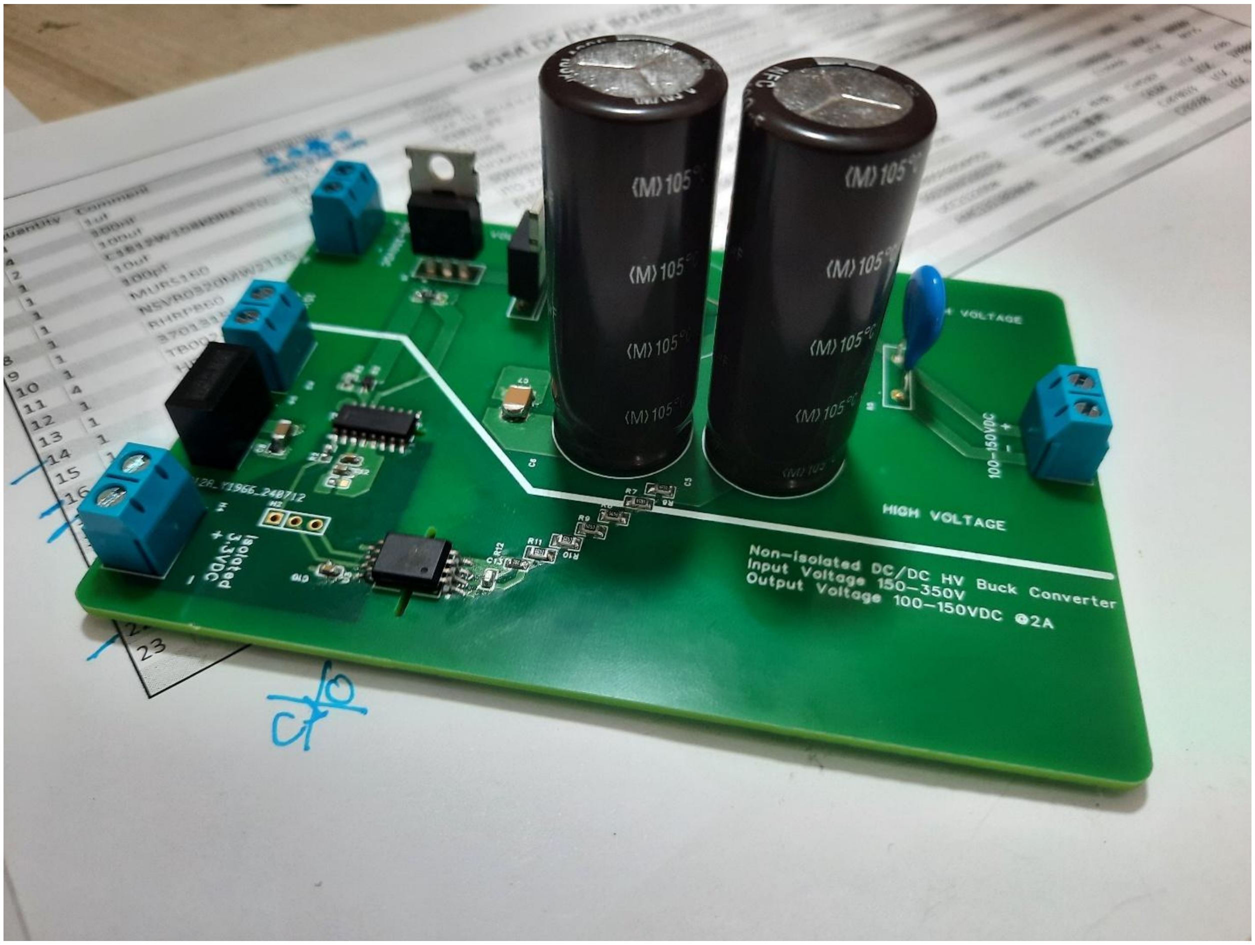
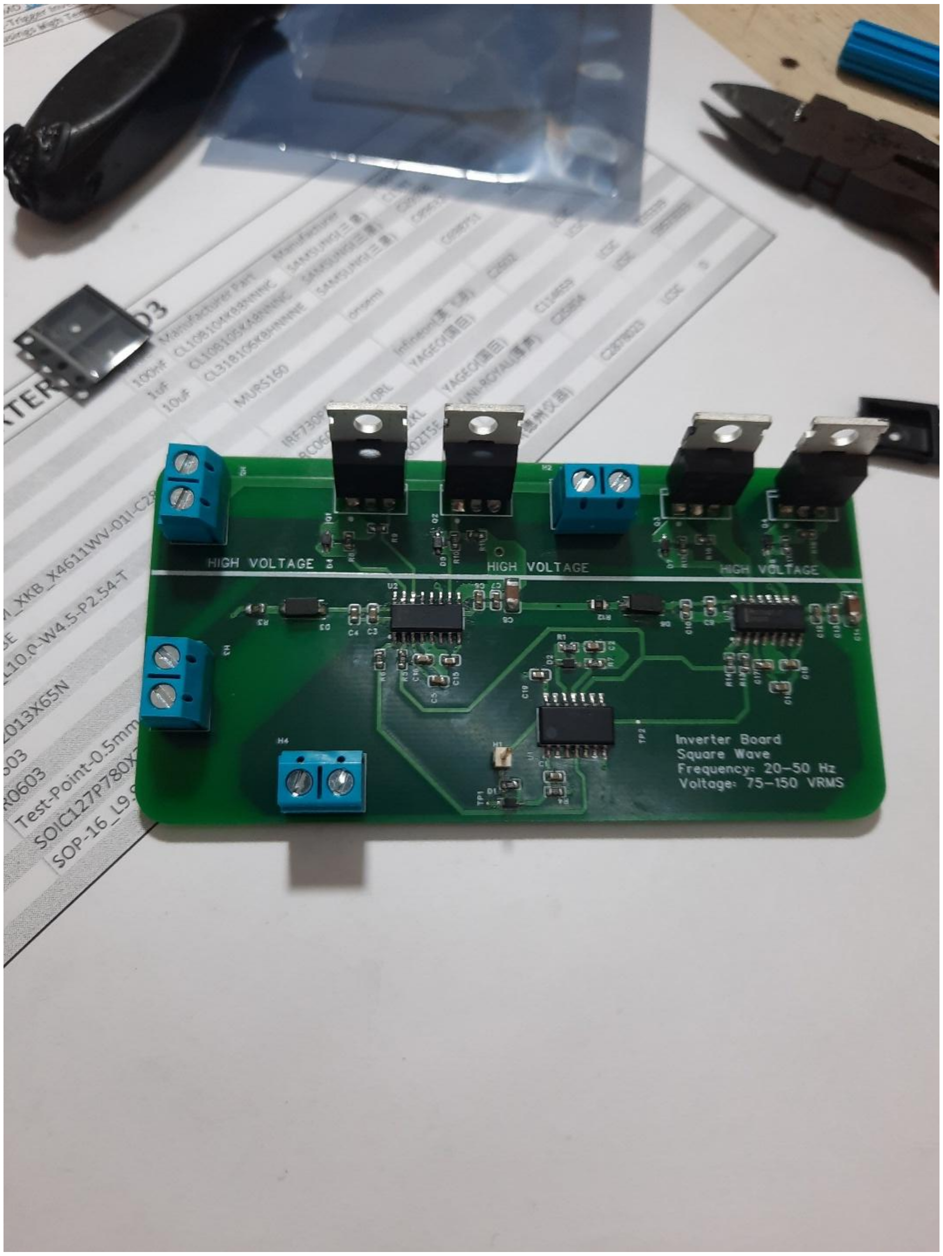


Figure 8 High Voltage Gate Driver IC

Figure 9 assembled buck converter and voltage sensing circuit for feedback


Non-Isolated DC/DC HV Buck Converter
Input Voltage 150-350V
Output Voltage 100-150VDC @2A
HIGH VOLTAGE
100-150VDC
Isolated
+3.3VDC

Assembled Inverter PCB




Inverter Board
Square Wave
Frequency: 20-50 Hz
Voltage: 75-150 VRMS
HIGH VOLTAGE
HIGH VOLTAGE
HIGH VOLTAGE

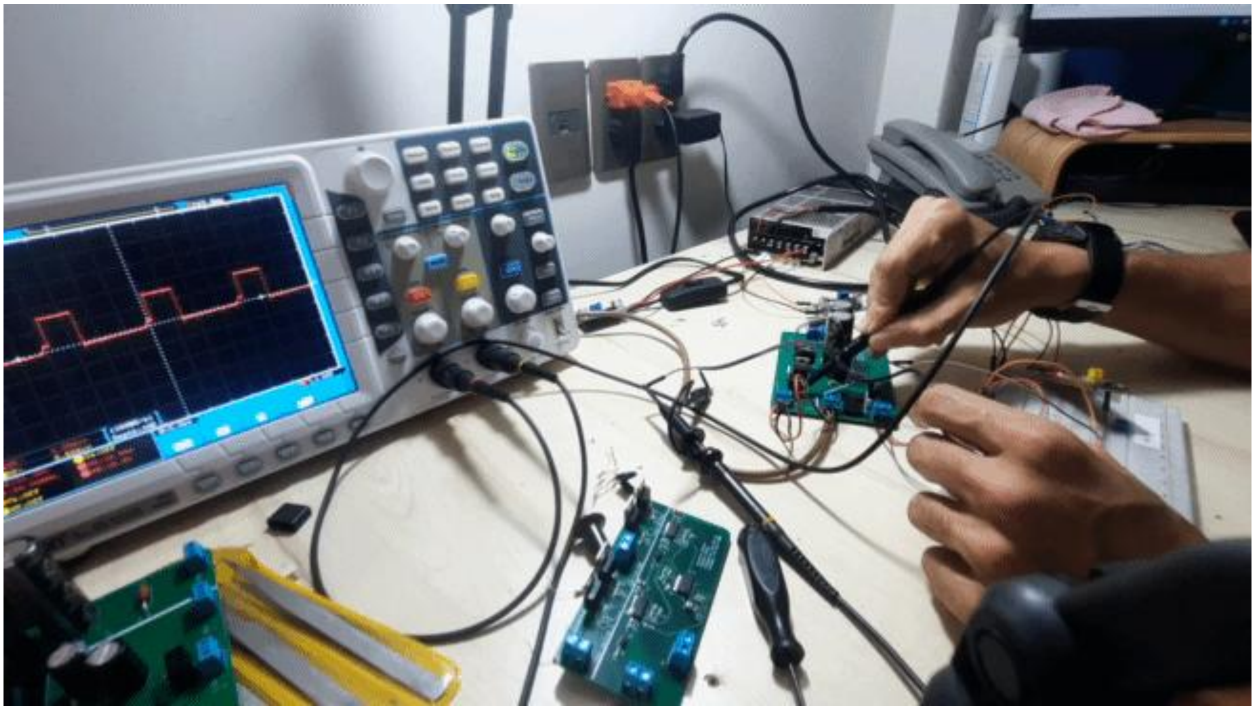
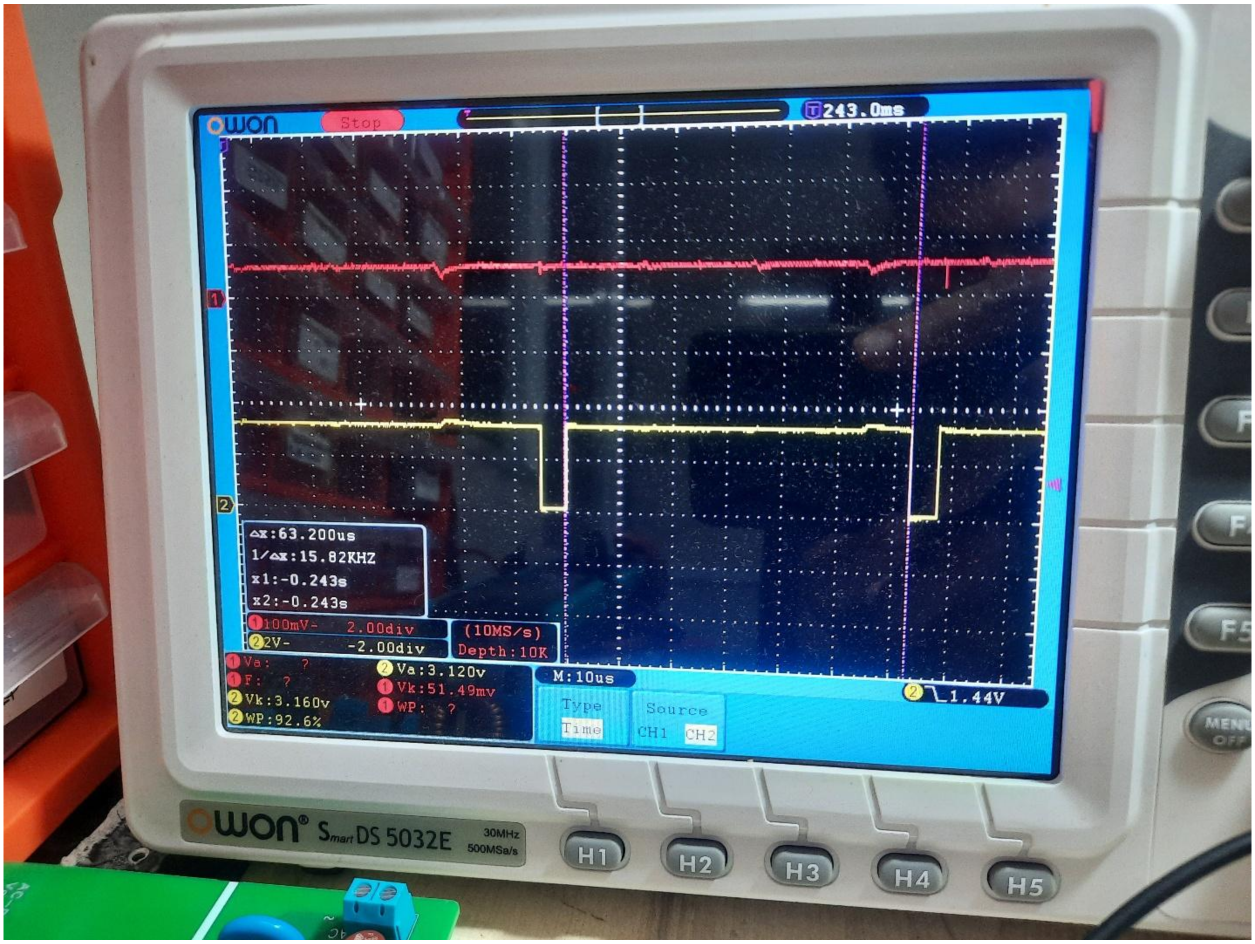
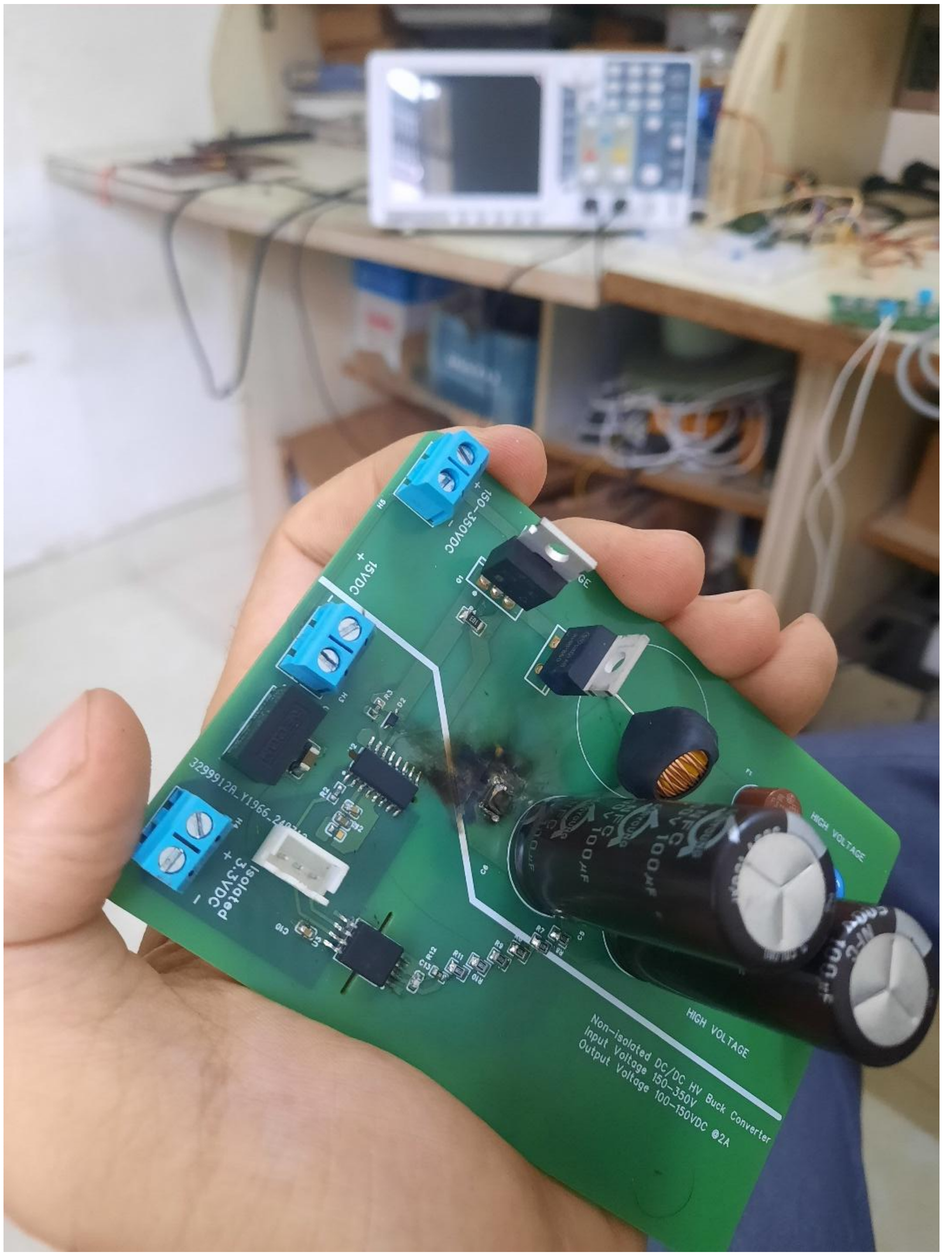


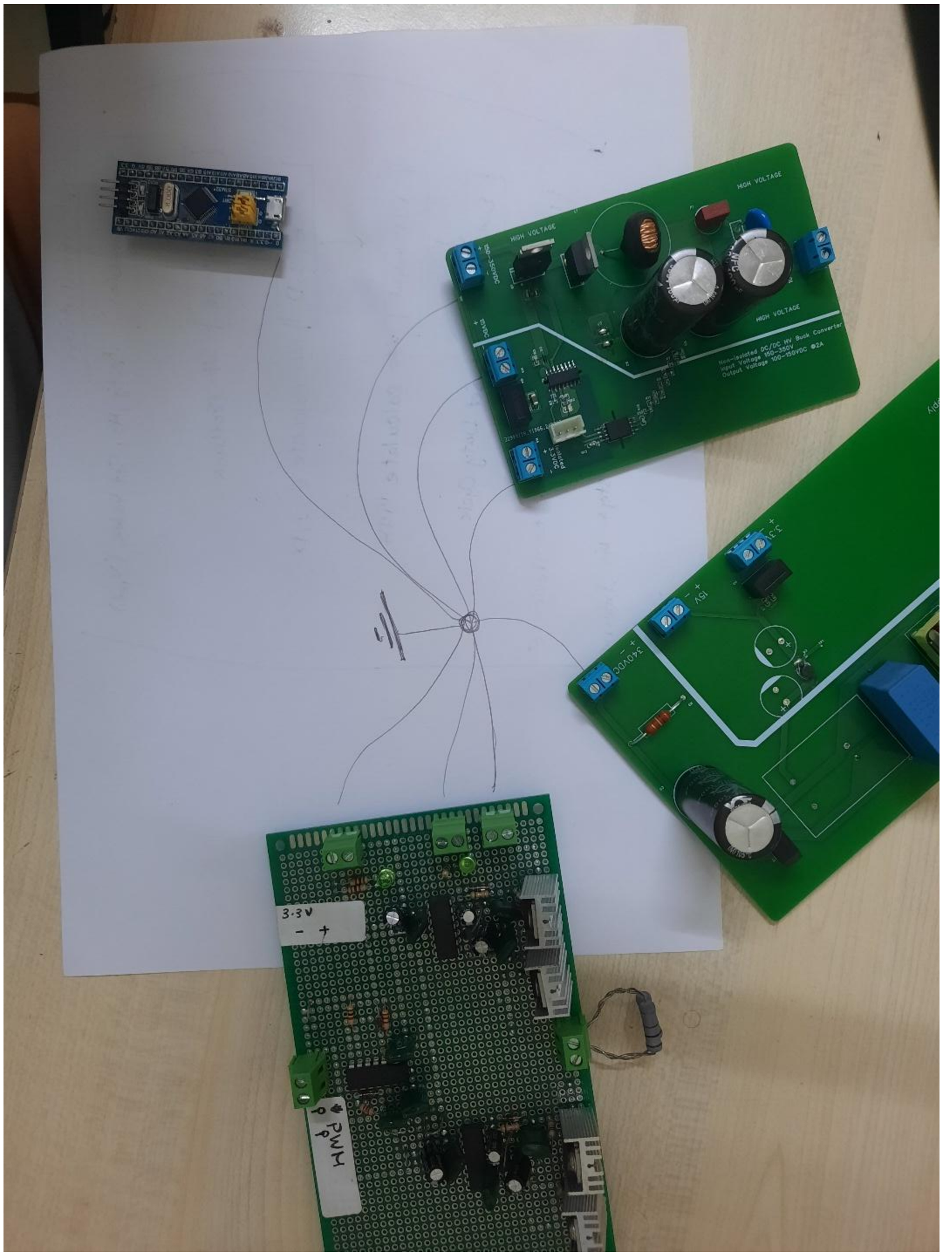
Figure 10 Testing signal integrity



OWON
Stop
243.0ms
△x:63.200us
1/△x:15.82KHZ
x1:-0.243s
x2:-0.243s
100mV- 2.00div
2V- -2.00div
(10MS/s)
Depth:10K
Va: ?
F: ?
Vk:3.160v
WP:92.6%
Va:3.120v
Vk:51.49mv
WP: ?
M:10us
Type
Time
Source
CH1
CH2
1.44V
OWON SmartDS 5032E
30MHz
500MSa/s
H1
H2
H3
H4
H5

Figure 11 exploded capacitor: Unexpected Lesson in High-Voltage Creepage and Clearance


150-350VDC
15VDC
Isolated
3.3VDC
HIGH VOLTAGE
HIGH VOLTAGE
Non-isolated DC/DC HV Buck Converter
Input Voltage 150-350V
Output Voltage 100-150VDC @2A


HIGH VOLTAGE
3.3V
- +
PWM

Simulation

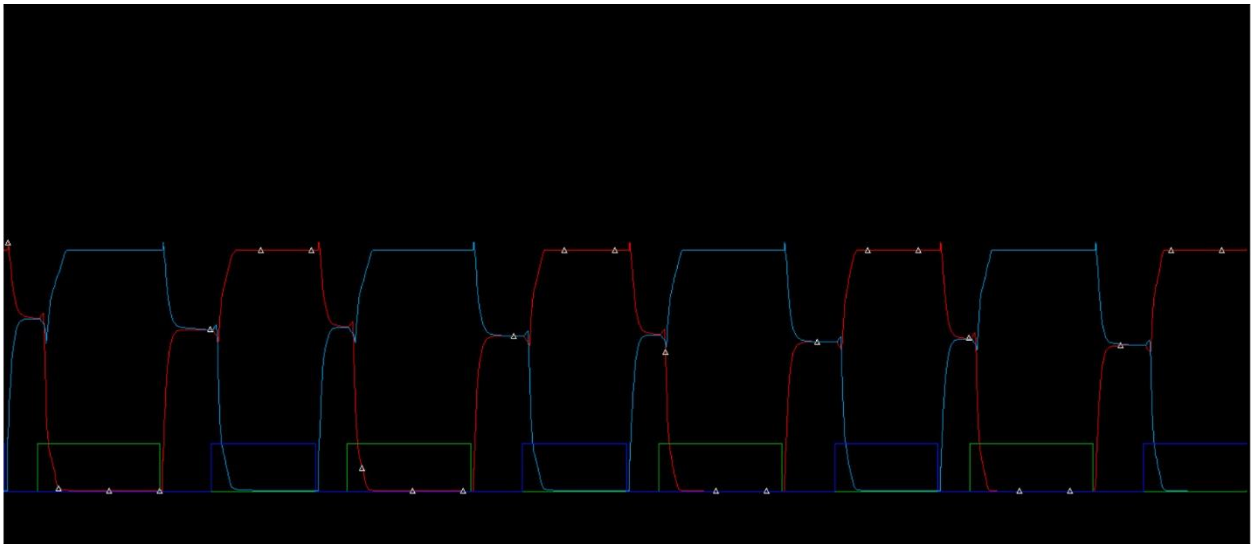


Figure 12 Deadtime circuit simulation on LTspice

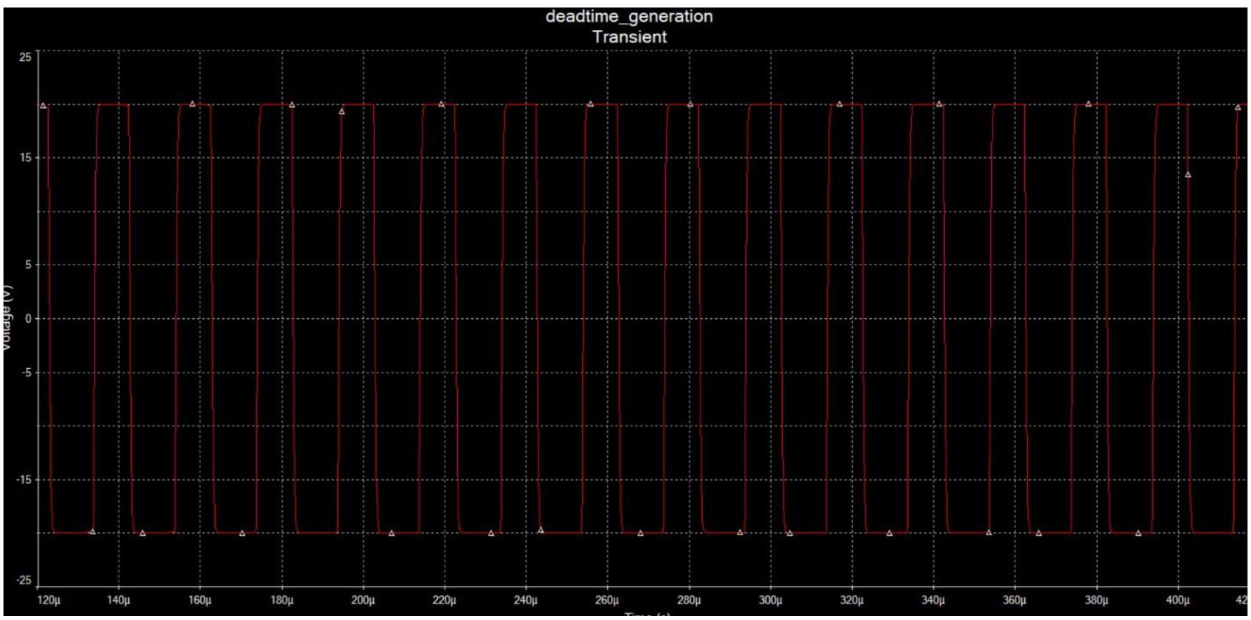


Figure 13 Deadtime generation circuit